

6.4.3 – 6.4.5 Prototypes, Observation, and Simulations

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جا سکے۔

CS XI

Chapter 6: Digital Literacy

Lecture 05 of 14

★ Golden

6.4.3 – 6.4.5 Prototypes, Observation, and Simulations ★ Golden

6.4.3 Prototypes

LEARN IT

A **prototype** is a **simple early version** of a product, model or solution. It allows an idea to be **tested before** time and effort are spent on the final version. Users **try** the prototype and give **feedback**, and this feedback becomes useful **data for improvement**.

Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five classmates to try "finding the timetable". You note where they get confused.

6.4.4 Observation

LEARN IT

In **observation**, the researcher **watches** what people do or what happens in a situation and **records the results carefully**. **No direct questions are asked**.

The data may be **qualitative** (written notes, e.g. "students push in the queue") or **quantitative** (counts, e.g. "42 students bought samosas in 10 minutes").

Why use it? People sometimes say one thing and do another. Observation shows what really happens.

6.4.5 Simulations

LEARN IT

A **simulation** is a **computer-based model of a real situation**. **Variables** (the things that can change, like speed, temperature or number of people) can be changed to see what happens.

Simulations are highly useful when **real experiments are costly, risky or difficult** to perform.

Examples: a flight simulator for training pilots (real crashes are dangerous); a traffic simulation to test what happens if a new signal is added (changing real roads is costly); a weather model to predict a cyclone (you cannot experiment with real weather).

WORKED EXAMPLE

Choosing the right method. Which method would you use?

(a) "How many cars cross an intersection at 8 AM?" → Observation. Stand at the corner from 8:00 to 8:15 and tally each car. Asking drivers would be impossible and unnecessary; you just count.

(b) "How do students feel about the new cafeteria food?" → Survey or interview. Feelings are inside people's heads, so you must ask them. Use a survey for many students, or interviews for deeper reasons.

(c) "Will a new college website menu be easy to use?" → Prototype. Build a rough version and let users try it.

(d) "What happens to traffic if the bridge is closed for repairs?" → Simulation. Closing a real bridge to test is costly and disruptive.

REMEMBER IT

Watch it → Observation. **Ask** it → Survey/Interview. **Try** it → Prototype. **Model** it → Simulation.

COMMON MISTAKE

Mistake: Saying that observation involves asking people questions. **Fix:** In observation, **no direct questions** are asked – the researcher only watches and records. Also, a prototype is *not* the finished product; it is an early test version.

اردو میں سمجھیں

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جا سکے۔ مشاہدے (Observation) میں محقق کوئی سوال نہیں پوچھتا بلکہ صرف دیکھ کر نتائج درج کرتا ہے۔ سمیولیشن کسی حقیقی صورتحال کا کمپیوٹر ماڈل ہے، جو اس وقت مفید ہے جب حقیقی تجربہ مہنگا، خطرناک یا مشکل ہو۔

EXAM TIP

Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method **and** justify it using its key feature (e.g. "Simulation, because a real experiment would be risky").

CHECK YOURSELF

1. Why would an engineer use a simulation instead of a real test?
2. Can observation produce quantitative data? Give an example.
3. What becomes the "data" when using a prototype?

Answers: 1. Because the real experiment may be costly, risky or difficult. 2. Yes – for example, counting the number of cars passing a point. 3. The feedback users give after trying the prototype.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A prototype here is:

- a. An early model to test an idea
- b. The final app store release
- c. A primary key
- d. A histogram

MCQ 2. Observation means:

- a. Watching what people actually do
- b. Only asking them
- c. Only simulating
- d. Only Googling

MCQ 3. Simulation is:

- a. A model of a system you can run
- b. A real flood in the campus
- c. A NAND chip
- d. A foreign key

MCQ 4. People say they wait 2 minutes but observation shows 10: trust:

- a. Observation of that behaviour
- b. The boast
- c. Neither ever
- d. OSI

MCQ 5. Paper screens of an app are a:

- a. Low-fidelity prototype
- b. MVP already selling
- c. Database schema
- d. Pie chart

MCQ 6. Simulations are useful when:

- a. Real experiments are costly or unsafe
- b. You already have all answers
- c. $n=1$ always
- d. Ethics forbid models

Extra short questions

1. Define prototype, observation, simulation in one line each.
2. One limit of observation.
3. One limit of simulation.
4. When is a prototype better than a survey?

Extra long question

1. A team wants a quieter library. Combine prototype, observation and simulation in a plan with ethics.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Prototype: early model. Observation: watch. Simulation: run a model.
2. Hawthorne effect — people act differently when watched.
3. Wrong assumptions → pretty but false results.
4. When you need to see if a design is usable, not just popular in words.

Long

1. Observe noise sources; prototype signage/zones; simulate seating. Consent to watch; no filming faces for TikTok. What would change your mind (data).

Next lecture: 6.5 Primary and Secondary Data